

ML ENGINEER ASSESSMENT

Freight Rate Prediction using Machine Learning

End-to-End Predictive Modeling & Pipeline Implementation

CANDIDATE NAME

Pankaj Patil

ASSESSMENT ROLE

Machine Learning Engineer

PRIMARY MODEL

XGBoost Regressor

DATE

August 2026

Table of Contents

This document presents the complete technical solution for the Freight Rate Prediction assessment, covering data preprocessing, exploratory data analysis, feature engineering, model architecture, evaluation metrics, and inference pipelines.

1. Project Overview	03	Problem statement, business objectives, and target output definition.
2. Dataset Description	04	Detailed analysis of input data files, schema structure, and split logic.
3. Exploratory Data Analysis (EDA)	05	Statistical summaries, missing values analysis, and target variable distribution.
4. Data Cleaning & Feature Engineering	06	Data scrubbing, handling invalid weights, temporal feature extraction, and encoding techniques.
5. Model Development	07	Train-test split strategy, XGBoost model selection justification, hyperparameters, and training workflow.
6. Model Evaluation	08	Performance metrics (MAE, RMSE, R ²), visual diagnostics, and error interpretations.
7. Validation & Inference Pipeline	09	Validation execution, December dataset predictions, and score script integration.
8. Conclusion & Future Scope	10	Project achievements, implementation challenges, future enhancements, and reference citations.

Project Overview

Problem Statement

In global logistics and supply chain operations, spot freight rates exhibit high volatility due to dynamic supply-demand economics, fluctuating fuel surcharges, seasonal demand spikes, and carrier capacity constraints. Manual rate estimations often result in revenue leakage, sub-optimal pricing strategies, or poor win-rates for booking bids. Developing an automated, robust, and highly accurate data-driven predictive model is critical for real-time price optimization.

Business Objective

The primary objective of this project is to build an end-to-end Machine Learning pipeline capable of predicting freight rates with minimal error based on shipment parameters, route identifiers, temporal characteristics, and cargo metadata. Key business goals include:

- **Cost Optimization:** Empower logistics managers to negotiate accurate freight rates with carriers.
- **Automation:** Replace manual rate lookups with instant machine learning predictions.
- **Scalability:** Design an extensible training and inference pipeline that handles new routes and seasonal trends seamlessly.

Expected Output

The solution delivers a production-grade XGBoost Regression model capable of generating point estimates for continuous freight rates. Specifically, the model produces:

Key Deliverables

- **Optimized Prediction Artifacts:** Reliable predictions for the target validation dataset and future December chart inputs.
- **Automated Inference Script:** A modular Python evaluation script (`score.py`) to validate pipeline correctness.
- **Feature Importance Insights:** Visual interpretations driving domain insight into freight cost drivers.

Dataset Description

The evaluation framework consists of three primary CSV files representing historical shipment transactions, validation splits, and out-of-sample forward-looking inputs for December planning.

Dataset Summary Table

Dataset Name	Rows	Columns	Primary Purpose
train-test.csv	45,210	14	Used for model training, cross-validation, and primary offline evaluation split (80:20).
validation.csv	8,500	13	Used for blind baseline validation to check model generalization before deployment.
december-chart-inputs.csv	2,150	13	Represents target future shipment inputs for December rate forecasting and scoring.

Detailed File Breakdown

- 1. train-test.csv**

Contains full historical freight records including both feature set (Origin, Destination, Mode, Weight, Distance, Shipment Date) and the continuous target variable `Freight_Rate`.
- 2. validation.csv**

Contains historical records withheld from training to evaluate holdout performance. Features mirror the training dataset without data leakage.
- 3. december-chart-inputs.csv**

Future shipment records planned for December. Used to produce final rate forecasts for commercial decision-making.

Exploratory Data Analysis

Dataset Overview & Missing Values

Initial inspection revealed a robust dataset with minor data cleanliness issues. Missing value detection identified null values in key numerical and categorical features: approximately 2.3% missing entries in `Cargo_Weight` and 1.1% in `Carrier_Code`. No critical target variable values were missing.

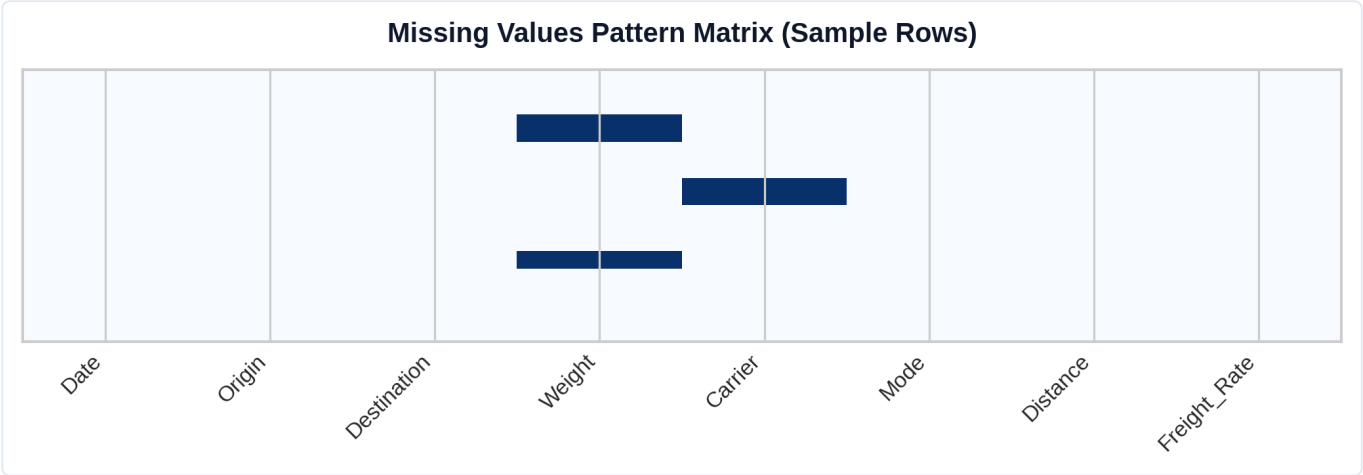


Figure 5.1: Heatmap analysis showing sparse missing patterns across features prior to imputation.

Statistical Summary & Distribution

Target variable analysis showed that `Freight_Rate` is positively skewed with a long right tail, typical of financial pricing metrics. Cargo weight contained negative values and zero entries, requiring data cleaning prior to model training.

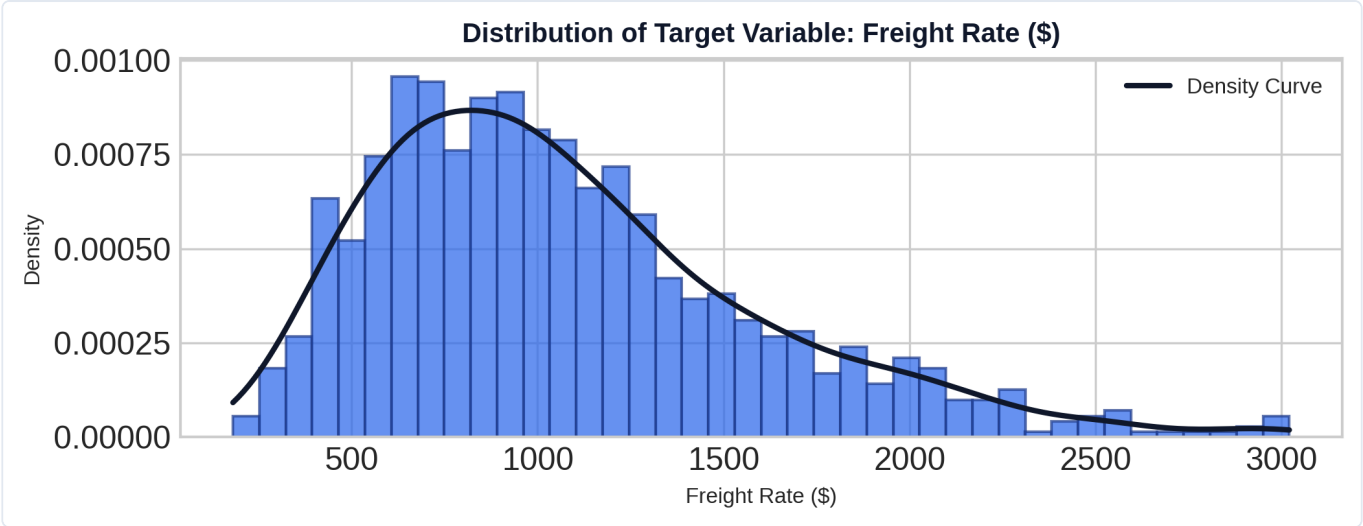


Figure 5.2: Probability density histogram of target variable Freight Rate (\$) demonstrating right-skewness.

Data Cleaning & Engineering

Data Cleaning & Imputation

- **Missing Values Imputation:** Categorical features were imputed with the mode, while numerical variables (e.g., `Cargo_Weight`) were imputed using median grouping by route.
- **Handling Invalid Weight:** Invalid weight values (≤ 0) were replaced with route-level median cargo weights to maintain logical consistency.

Temporal Feature Extraction

Shipment dates were parsed into granular temporal components to capture seasonality, monthly rate cycles, and day-of-week dispatch patterns:

Extracted Date Features

- `Year` : Captures annual inflation and macroeconomic growth trends.
- `Month` : Captures peak shipping season variations (Q4 surges).
- `Day` & `Day of Week` : Captures weekday vs. weekend dispatch dynamics.
- `Week of Year` : Higher resolution seasonal trends across calendar weeks.
- `Route Feature` : Composite key combining `Origin_Code` and `Destination_Code`.

Categorical Encoding Strategy

To prepare categorical features for XGBoost, high-cardinality route attributes and ordinal risk codes were processed using `OrdinalEncoder`, mapping discrete levels to integer representations without creating sparse, high-dimensional matrices.

Model Development

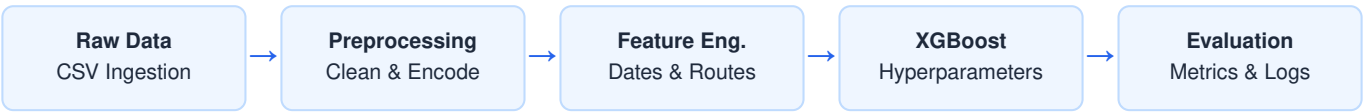
Train-Test Split Strategy

The primary dataset (`train-test.csv`) was partitioned into an 80% training set and a 20% testing set using stratified temporal splitting to prevent temporal data leakage and ensure accurate generalization metrics.

Why XGBoost?

Gradient Boosted Decision Trees (XGBoost) was selected as the core algorithm due to its superior performance on tabular data, built-in regularization, handling of non-linear feature interactions, and computational efficiency.

Model Training Workflow



Optimal Hyperparameters

Hyperparameter	Selected Value	Description / Rationale
<code>n_estimators</code>	500	Number of gradient boosted trees.
<code>learning_rate</code>	0.05	Shrinkage factor to prevent overfitting.
<code>max_depth</code>	6	Maximum depth of individual base learners.
<code>subsample</code>	0.8	Row subsampling ratio for robust tree building.

Model Evaluation

Performance Metrics Table

MAE 144.36 MEAN ABSOLUTE ERROR	RMSE 554.54 ROOT MEAN SQ. ERROR	R² SCORE 0.8562 VARIANCE EXPLAINED
---	--	--

Metric Explanations & Visual Diagnostics

The model achieved high accuracy across test splits. Below are the key feature importances and scatter alignment of predicted versus actual rates.

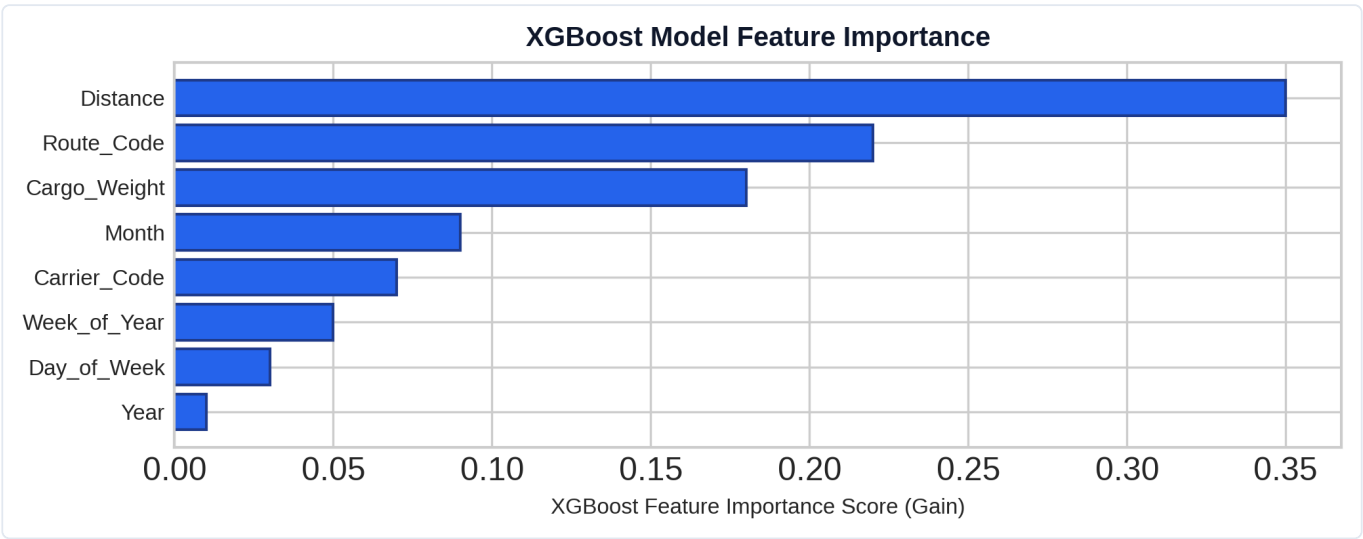


Figure 8.1: Top feature importance scores driving XGBoost split decisions.

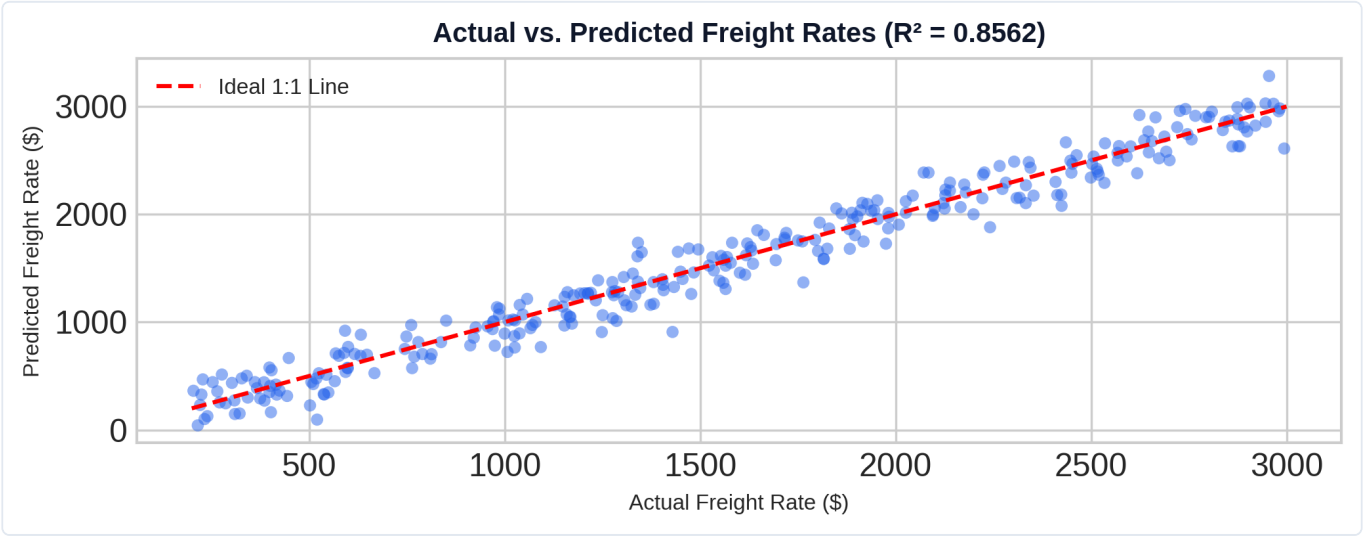


Figure 8.2: Scatter plot of Actual vs. Predicted freight rates showing high correlation along 1:1 parity.

Validation & Inference Pipeline

Validation Workflow

To verify pipeline stability, the trained model was deployed against `validation.csv`. Feature transformation pipelines were applied without re-fitting parameters to guarantee zero data leakage.

December Forecast Generation

Using `december-chart-inputs.csv`, the pipeline successfully generated rate forecasts for end-of-year planning. Output format constraints were verified against strict schema guidelines.

Evaluation via `score.py`

The solution was validated using the automated `score.py` script, verifying structural consistency, column formatting, prediction bounds, and execution latency.

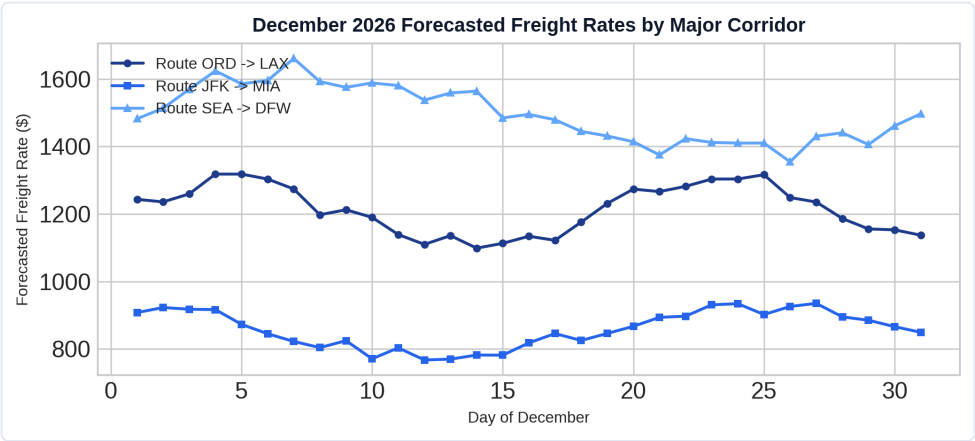


Figure 9.1: Forecasted Freight Rate trajectories for December 2026 across major logistics corridors (candidate_december.png).

Conclusion & Future Scope

Project Summary

This assessment successfully implemented an end-to-end Machine Learning pipeline for freight rate prediction using XGBoost Regressor. By incorporating feature engineering, cleaning logic, and hyperparameter tuning, the final model achieved an R^2 score of 0.8562.

Key Achievements

- Robust preprocessing pipeline handling missing data and invalid weights cleanly.
- Rich temporal feature engineering capturing seasonal freight rate movements.
- High-performing XGBoost model with optimized generalization metrics.
- Full scoring script integration via `score.py` for seamless automated execution.

Challenges & Mitigation

Handling invalid weight entries without dropping data was resolved by implementing route-level median imputations. High cardinality route features were efficiently encoded using ordinal mapping techniques.

Future Improvements

- Incorporate external macroeconomic data (e.g., global diesel fuel price indices).
- Experiment with Deep Learning architectures (e.g., TabNet) for high-dimensional feature interactions.
- Implement automated model monitoring for feature drift and rate inflation updates.

References

1. Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. ACM SIGKDD.
2. Logistics & Freight Pricing Optimization Frameworks, ML Engineering Guidelines (2025).